

Final Project Instruction: Platform-Based Management System with Cloud Backend

Objective

Design and develop a **platform-based management system** that runs on a web platform using modern frontend technologies and a cloud backend service such as Firebase or similar.

Project Description

Students will create a **functional management system platform** that demonstrates:

- Interactive user interface (forms, dashboards, navigation)
- Data management (create, update, delete, display records)
- Cloud-based backend integration
- Optional user authentication and access control
- Responsive and user-friendly design

Technical Requirements

1. Frontend (Choose Any)

You may use:

- HTML, CSS, JavaScript (Vanilla)
- React
- Vue.js
- Next.js
- Bootstrap or Tailwind CSS (optional)

2. Backend (Required)

Use **any cloud backend service**, such as:

- Firebase (Firestore or Realtime Database)
- Supabase (alternative)

- Appwrite (alternative)

Backend must support:

- Data storage and retrieval
- API or SDK integration with frontend
- Optional authentication system

Minimum System Features

1. User Interface

- Clean, organized, and responsive layout
- Navigation system (sidebar, header, or menu)
- Dashboard-style interface (recommended)

2. CRUD Operations

Your system must allow:

- Create records
- Read/display records
- Update/edit records
- Delete records

3. Cloud Integration

- Connect to backend database
- Store and retrieve real-time or dynamic data

4. Form Handling

- Input validation
- Proper submission and processing

5. Optional Features (Bonus)

- User Authentication (Login/Register)
- Role-based access (Admin/User)
- Real-time updates
- File uploads (images/documents)
- Notifications or alerts

Suggested Platform-Based System Ideas

- Student Management System
- Hotel/Resort Reservation System
- Inventory Management System
- Appointment Booking System
- Task Management Platform
- Online Ordering System

Deployment Options

You may deploy your project using:

- **Vercel** (recommended)
- ****Firebase Hosting**
- **Netlify** (alternative)

Grading Criteria (100 Points)

Criteria	Description	Points
Functionality	System works correctly, minimal bugs	30
Backend Integration	Proper use of Firebase or alternative	20
UI/UX Design	Clean, responsive, user-friendly	15
Code Quality	Organized and readable code	15
CRUD Implementation	Complete CRUD features	10
Creativity & Features	Unique idea, extra features	5
Presentation / Demo	Clear explanation and demo	5

Submission Requirements

- Source Code (ZIP or GitHub repository)
- Documentation (PDF or Word):
 - Project Title
 - Description
 - Technologies Used
 - Features
- Live Demo Link (if deployed)

Deployment Guide (Using Vercel)

1. Upload your project to GitHub
2. Sign in to **Vercel** using GitHub
3. Import your repository
4. Configure (if needed):
 - Build Command: `npm run build`
 - Output Folder: `dist` or `build`
5. Click **Deploy**
6. Get your live URL

Important Notes

- Ensure backend configuration is correct
- Avoid exposing sensitive credentials
- For authentication:
 - Add your deployment domain in backend settings
- Test your system before submission

Deadline

Project Submission Window: May 20 – May 23

- Late submissions may incur penalties unless approved

